Name: _____

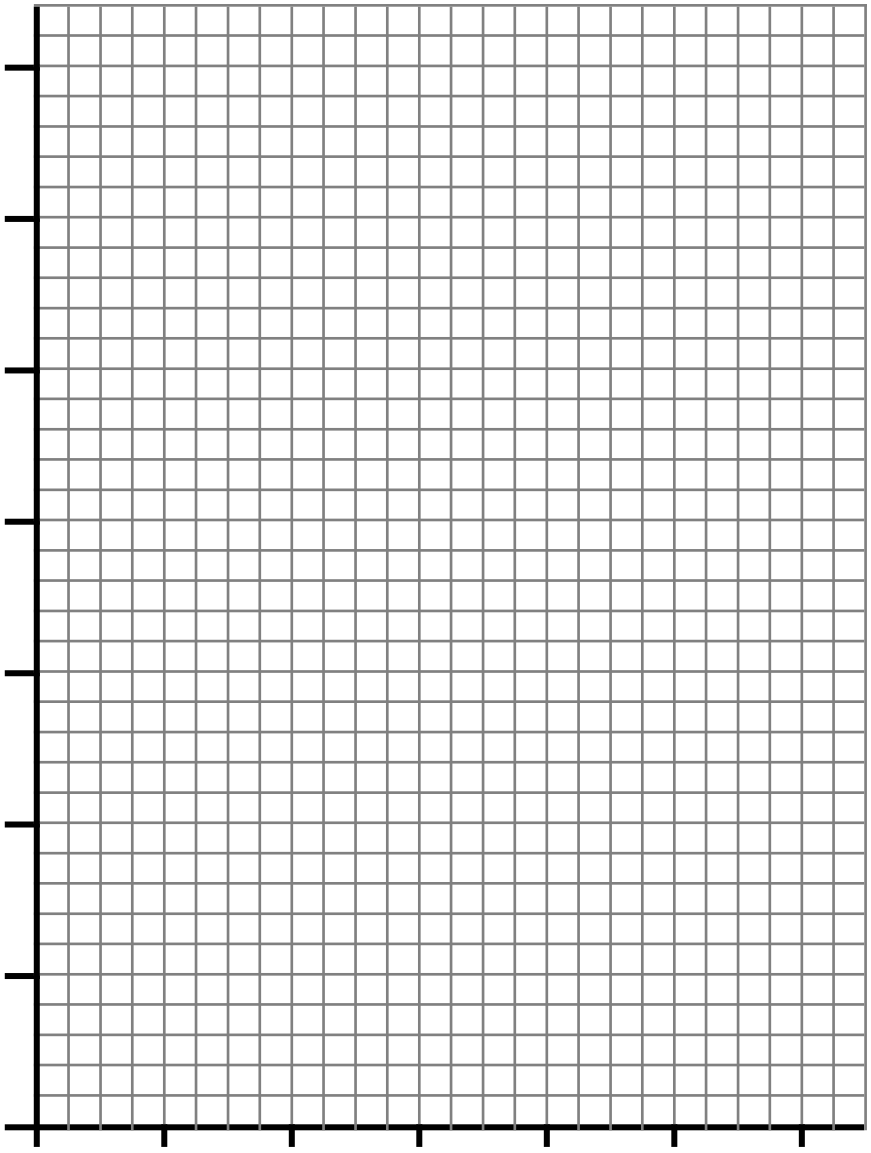
Specific Heat Capacity

Activity 1: Heating Substances in the Sun

The following table shows the temperature after 10g of four different substances have been in direct sunlight for 60 minutes.

Time (mins)	Air (ºC)	Water (ºC)	Sand (ºC)	Metal (ºC)
0	25	25	25	25
15	29	26	30	35
30	33	28	35	45
45	36	29	40	55
60	40	30	45	65

Create a line graph for each substance on the grid below. Put the time on the x-axis (along the bottom) and the temperature on the y-axis (up the side). Don't forget to give your graph a title.



1. Order the substances based on the time required to heat them from slowest to fastest.

2. Which do you think will cool the fastest? Explain your choice.

3. When you boil water in a pot on the stove, which heats up faster, the metal or the water? Explain your choice.

4. The specific heat capacity is the amount of heat needed to raise the temperature of 1g of a substance by 1°C. A substance with a high specific heat capacity heats up more slowly than one with a low specific heat capacity because it needs to absorb more energy before it heats up.
   a. Which of the four substances has the highest heat capacity? Explain your choice.

   b. Which of the four substances has the lowest heat capacity? Explain your choice.

5. The specific heat capacities of the four substances are **4200**J/kg/^{o}C, **880**J/kg/^{o}C, **1000**J/kg/^{o}C and **400**J/kg/^{o}C. Match each specific heat capacity to one of the substances.
   a. Air = _____
   b. Water = _____
   c. Sand = _____
   d. Metal = _____

6. Assuming they both start at the same temperature, which will heat up faster in the sun, a bottle of water or a swimming pool full of water? Explain your choice.

Activity 2: Calculate the amount of heat absorbed or released (show all working)

1. A pot is filled with 9kg of water at 10°C. Calculate how much heat energy would be needed to raise the temperature to 60°C. [specific heat capacity of water = 4200J/kg/^{o}C]

2. A girl is hiking in the bush and has a 2litre bottle of water, which weighs 2kg, in her backpack. The water gets heated by the sun and its temperature increases by 5°C. How much energy has it absorbed?

3. If the temperature of 3.5kg of water increases from 50°C to 100°C, how much water has been absorbed by the water?

4. How much energy is needed to raise the temperature of a 2kg bar of gold from 25°C to 85°C? [specific heat capacity of gold = 130J/kg/^{o}C]

5. Half a kilogram of water at room temperature (25°C) is put in the freezer until it freezes at 0°C. How much heat energy has it released?

6. An iron object weighing 4kg was accidentally left outside in the sun. When it was brought inside, its temperature was 45°C. How much energy will it lose once it has cooled to room temperature? [specific heat capacity of iron = 440J/kg/^{o}C]